

Project 1 — Milestone

Phase A · before data collection · Module 1

Project 1 milestone

Phase A in : Project 1 — Testing a Claim

! Milestone notebook

[project-1-milestone.ipynb](#) · Colab

Complete **Phase A** (steps **A1–A5**) from **Project 1** in the notebook, then submit in the Canvas for early feedback on your project.

What this checkpoint is for

This check allows you plan and defend your study before collecting all $n = 25$ trials (Phase A on the **Project 1** page). That way we catch a bad process, vague p , or mismatched hypotheses early. Do not collect data, record x , mark your data on the null plot, or compute a p-value yet — those are Phase B.

Why state H_0 and H_a before you collect data?

In **Module 0, Read 0**, the five-step statistical process puts planning before collecting. Step 2 is to decide what you will collect, how you will analyze it, and how you will conclude — *before* Step 3 (gather data from the world we observe). Phase A of this milestone is your Step 2 for Project 1: you fix the research question, define p , and write H_0 and H_a with a numeric null p so your later analysis is decided in advance, not chosen after you have seen x .

Module 1, Read 0 adds the three worlds. Your future counts (x out of $n = 25$) will live in the world you observe. The world we imagine is where you *pretend* a specific p is true and ask what counts would be typical — that is exactly what H_0 specifies. Your null binomial plot (step A5) is a picture of that imagined world under H_0 , drawn before you know whether your real x will look surprising. If you wait until after data collection to pick H_0 , you blur the line between “what the model predicts if the claim were true” and “what I already saw,” and the p-value loses its meaning as evidence against a claim stated in advance.

Examples: research question $\rightarrow H_0$ and H_a

Use these as patterns, not as topics to copy. Your Project 1 process must be your own ($n = 25$, not ESP or other class demos). In each case, H_0 uses an equality and H_a matches the direction of your research question ($>$, $<$, or $\neq$). More worked examples appear in **Read 2**.

Example 1

- Research question: Is the proportion of left-handed students at our school higher than the national rate of 10%?
- p in context: probability that a randomly chosen student at our school is left-handed.
- Best guess about p : probably above 0.10.
- Hypotheses: $H_0 : p = 0.10$ vs. $H_a : p > 0.10$

Example 2

- Research question: Is the proportion of defective items from a supplier less than their claimed rate of 15%?
- p in context: probability that one item from the supplier is defective.
- Best guess about p : probably below 0.15.
- Hypotheses: $H_0 : p = 0.15$ vs. $H_a : p < 0.15$

Example 3

- Research question: Is the germination rate for these seeds different from the 80% rate printed on the package?
- p in context: probability that one seed germinates under your growing conditions.
- Best guess about p : might be above or below 0.80; either direction would be interesting.
- Hypotheses: $H_0 : p = 0.80$ vs. $H_a : p \neq 0.80$

i Note

If your research question is one-sided in plain language (“higher than,” “less than”), use H_a with $>$ or $<$. In practice, most studies use the two-sided $H_a : p \neq p_0$.

Phase A — complete in the notebook

Step	Task
A1	Choose a binomial process with $n = 25$ (not a class example). Check all four BRP conditions: (a) binary outcome per trial, (b) independent trials, (c) fixed p per trial, (d) fixed n .
A2	Write your research question with your best guess about the size of p in the infinite process.
A3	Define p in context — what counts as one “success” on a single trial.
A4	State H_0 and H_a with a numeric p ($0 < p < 1$). Write H_a by translating your research question into our notation.
A5	Plot the null binomial distribution in R (<code>dbinom + geom_col</code>) using $n = 25$ and p from H_0 . Do not mark an observed x yet.

What to turn in to Canvas

Submit the notebook (or exports your course asks for) showing your work for A1–A5, especially:

1. Your process and why each BRP condition holds in context (A1)
2. Research question and your guess about p (A2)
3. One sentence defining success / p (A3)
4. H_0 and H_a with a numeric null p (A4)
5. Null binomial graph only — no observed x , no p-value (A5)

Code help

- [Module 1 demo lab](#) — `dbinom + geom_col`

After the milestone

Complete **Phase B** on the [Project 1](#) page:

1. Observe at least $n = 25$ trials; record x and $\hat{p} = x/n$.
2. Descriptive bar or pie chart for observed counts.
3. Mark observed x on your null plot (`geom_vline`).
4. P-value with `pbinom()` (match H_a); interpret in context.
5. One-page summary using the [Project 1 rubric](#) (40 points total: 35 summary + 5 .ipynb).